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Economics and Data Analysis

Econometric Analysis of Social Well-Being: A Linear-Log and Log-Log Model Based on the Health and Retirement Study using R

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Course:
Econometrics

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Dataset:
no. 1

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Introduction

The data in the following analysis are an extract from the *Health and Retirement Study*, a longitudinal study that interviews a representative sample of U.S. people aged 50 years or older every two years. The data cover waves from 2006 to 2020 (8 waves). I will analyze the dataset *hrs_1*.

I start the analysis by running an OLS regression. It is a linear-log model that has *self-assessment of social status* as the dependent variable and a set of explanatory variables, in which there are some dummies and the logarithm of *income* and *financial wealth* expressed in USD. The OLS regression is represented by the following equation:

$$socladder = \beta_1 + \beta_2 age + \beta_3 age^2 + \beta_4 female + \beta_5 hi_school + \beta_6 college + \beta_7 empl + \beta_8 self empl + \beta_9 \log(income) + \beta_{10} \log(tot_finwth) + \beta_{11} wave_9 + \beta_{12} wave_10 + \beta_{13} wave_11 + \beta_{14} wave_12 + \beta_{15} wave_13 + +\beta_{16} wave_14 + \beta_{17} wave_15 + \epsilon$$

Since the relationship between the variable *age* and the dependent variable does not follow a linear trend, I decided to include its quadratic value in the specification of the model.

Regression Analysis of Income and Education Effects on Social Status

Point a

Report the effect of increasing income by 10%. Test if the effect is significantly different from zero.

Since this is a linear-log model, semi-elasticity is involved. It is possible to know the effect of a 10% increase in income by running on R the OLS regression, using the linear model command *(lm)*. I called this first regression *reg1*, including all the variables in the equation represented on the previous page. The results are as follows:

```
Call:
lm(formula = socladder ~ age + I(age^2) + female + hi_school +
    college + empl + selfempl + log(income) + log(tot_finw) +
    wave_9 + wave_10 + wave_11 + wave_12 + wave_13 + wave_14 +
    wave_15, data = dataset)

Residuals:
    Min      1Q  Median      3Q     Max
-6.6321 -0.9615  0.1751  1.1183  6.7681

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.8740537  0.5856940   1.492 0.135631
age            0.0627049  0.0164614   3.809 0.000140 ***
I(age^2)      -0.0003209  0.0001159  -2.768 0.005641 **
female        -0.0920870  0.0268274  -3.433 0.000599 ***
hi_school      0.2507011  0.0375099   6.684 2.41e-11 ***
college        0.8988051  0.0448428  20.043  < 2e-16 ***
empl           0.1192848  0.0351285   3.396 0.000686 ***
selfempl       0.3681307  0.0519173   7.091 1.39e-12 ***
log(income)    0.1685745  0.0107055  15.747  < 2e-16 ***
log(tot_finw)  0.0740797  0.0029669  24.968  < 2e-16 ***
wave_9TRUE    -0.0862099  0.0483732  -1.782 0.074739 .
wave_10TRUE   -0.1400707  0.0470546  -2.977 0.002918 **
wave_11TRUE   -0.1760483  0.0483925  -3.638 0.000276 ***
wave_12TRUE   -0.1141387  0.0479346  -2.381 0.017272 *
wave_13TRUE   -0.1559985  0.0525076  -2.971 0.002973 **
wave_14TRUE    0.1385308  0.0541079   2.560 0.010469 *
wave_15TRUE    0.1389335  0.0583860   2.380 0.017345 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.608 on 15229 degrees of freedom
Multiple R-squared:  0.1793,    Adjusted R-squared:  0.1785
F-statistic:   208 on 16 and 15229 DF,  p-value: < 2.2e-16
```

Figure 1: OLS regression with linear-log model.

It can be seen that if *income* increases by 100%, the dependent variable *self-assessed social status* increases by 0.168. Thus, a 10% increase in *income*, causes *self-assessed social status* to increase by 0.0168.

The coefficient is significant because it is marked with the symbol “***”. In any case, you can check its significance by running a t-test.

The set of hypotheses is as follows:

$$\begin{cases} H_0 : \beta_9 = 0 \\ H_1 : \beta_9 \neq 0 \end{cases}$$

To check its signficance I computed a t-test:

$$t - test = \frac{0.1685745}{0.0107055} = 15.747 > 1.96$$

Since the value is greater than 1.96, I reject the null hypothesis and conclude that the coefficient is significant. The value obtained is confirmed by the table above, in *Figure 2*, in the third column. Moreover, the p-value is less than *alpha* (5%), so I conclude that there is an income effect.

Point b

Test if the effects of high school and college degrees are the same.

To test whether the effects of having a high school diploma and a college degree are the same, I perform an F-test. The null hypothesis and the alternative hypothesis are set as follows:

$$\begin{cases} H_0 : \beta_5 = \beta_6 \Rightarrow \beta_5 - \beta_6 = 0 \Rightarrow \theta = 0 \\ H_1 : \beta_5 \neq \beta_6 \Rightarrow \beta_5 - \beta_6 \neq 0 \Rightarrow \theta \neq 0 \end{cases}$$

Since the hypothesis is a combination of coefficients, it is possible to manipulate the data so that there is only one value θ and not a combination of two. I run the linear hypothesis test command *lht* in R. The results are as follows:

```
Linear hypothesis test

Hypothesis:
hi_school - college = 0

Model 1: restricted model
Model 2: socladder ~ age + I(age^2) + female + hi_school + college + empl +
    selfempl + log(income) + log(tot_finw) + wave_9 + wave_10 +
    wave_11 + wave_12 + wave_13 + wave_14 + wave_15

  Res.Df   RSS Df Sum of Sq  Chisq Pr(>Chisq)
1  15230 40416
2  15229 39360  1    1055.5 408.39  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 2: F-test.

The test follows a Chi-squared distribution of parameter 1, whose critical value is 3.841. Since $408.39 > 3.841$ and the p-value is less than *alpha* (5%), I reject the null hypothesis and conclude that the effect of high school education and college degree are not the same. The effects are different.

Point c

Test if there is no joint effect of the wave.

It is possible to test the joint significance of the variable *wave* by running an F-test on R with the command *lht*. In this case, I'm dealing with multiple restrictions. Below I report the set of hypotheses and the result:

$$\begin{cases} H_0 : \beta_{11} = \beta_{12} = \beta_{13} = \beta_{14} = \beta_{15} = \beta_{16} = \beta_{17} = 0 \\ H_1 : \beta_{11}, \beta_{12}, \beta_{13}, \beta_{14}, \beta_{15}, \beta_{16}, \beta_{17} \neq 0 \end{cases}$$

```
Linear hypothesis test

Hypothesis:
wave_9TRUE = 0
wave_10TRUE = 0
wave_11TRUE = 0
wave_12TRUE = 0
wave_13TRUE = 0
wave_14TRUE = 0
wave_15TRUE = 0

Model 1: restricted model
Model 2: socladder ~ age + I(age^2) + female + hi_school + college + empl +
    selfempl + log(income) + log(tot_finw) + wave_9 + wave_10 +
    wave_11 + wave_12 + wave_13 + wave_14 + wave_15

  Res.Df   RSS Df Sum of Sq  Chisq Pr(>Chisq)
1  15236 39533
2  15229 39360  7       173 66.938  6.128e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 3: F-test on linear-log model.

The test follows a Chi-squared distribution of parameter 7, whose critical value is 14.067. Since $66.938 > 14.067$ and the p-value is less than 5%, I reject the null hypothesis and conclude that the coefficients are jointly significant.

Point d

Repeat your answer on a) from a model taking the log transformation of the dependent variable.

I need to run a new regression in which the dependent variable *self-assessment of social status* is a log-transformation. Below are the results of this new regression, which I named *reg2*:

```
Call:
lm(formula = log(socladder) ~ age + I(age^2) + female + hi_school +
    college + empl + selfempl + log(income) + log(tot_finw) +
    wave_9 + wave_10 + wave_11 + wave_12 + wave_13 + wave_14 +
    wave_15, data = dataset)

Residuals:
    Min      1Q  Median      3Q     Max
-2.0171 -0.1205  0.0563  0.1922  1.1438

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    5.676e-01  1.185e-01   4.792 1.67e-06 ***
age            1.552e-02  3.330e-03   4.661 3.17e-06 ***
I(age^2)      -8.368e-05  2.344e-05  -3.569 0.000359 ***
female        -1.226e-02  5.426e-03  -2.260 0.023845 *
hi_school      6.134e-02  7.587e-03   8.085 6.65e-16 ***
college        1.611e-01  9.070e-03  17.760  < 2e-16 ***
empl           4.295e-02  7.105e-03   6.046 1.52e-09 ***
selfempl       7.371e-02  1.050e-02   7.019 2.33e-12 ***
log(income)    3.408e-02  2.165e-03  15.740  < 2e-16 ***
log(tot_finw)  1.478e-02  6.001e-04  24.624  < 2e-16 ***
wave_9TRUE    -1.072e-02  9.784e-03  -1.096 0.273311
wave_10TRUE   -2.147e-02  9.517e-03  -2.256 0.024078 *
wave_11TRUE   -2.468e-02  9.788e-03  -2.522 0.011688 *
wave_12TRUE   -1.657e-02  9.695e-03  -1.709 0.087473 .
wave_13TRUE   -2.392e-02  1.062e-02  -2.252 0.024334 *
wave_14TRUE    2.873e-02  1.094e-02   2.625 0.008666 **
wave_15TRUE    2.798e-02  1.181e-02   2.369 0.017844 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3252 on 15229 degrees of freedom
Multiple R-squared:  0.165,     Adjusted R-squared:  0.1642
F-statistic: 188.1 on 16 and 15229 DF,  p-value: < 2.2e-16
```

Figure 4: OLS regression with log-log model .

The model becomes a log-log model because of the log transformation of the dependent variable. An increase in *income* by 10% creates an increase in *self-assessed social status* by 0.34%. The coefficient of the variable *income* is still significant.

Diagnostic Tests and Model Refinements

Point a

Perform a Breusch-Pagan test to check if the model exhibits homoskedasticity.

As requested, I will consider the baseline OLS regression taking as dependent variable the log transformation of *socladder*.

Homoskedasticity is one of the Gauss-Markov assumptions of OLS. It can be tested with White test and Breusch-Pagan (BP) test. As requested, I will proceed with the BP test, although it only verifies a specific case of heteroskedasticity. White test is more general. Below is the set of hypotheses:

$$\begin{cases} H_0 : \sigma_i^2 = \sigma^2 \\ H_1 : \sigma_i^2 = \sigma^2 h(z_i'\alpha) \end{cases}$$

Since we are considering the dependent variable subjected to log-transformation, I use the command *bptest(reg2)*. Below is the result of the test:

```
        studentized Breusch-Pagan test

data:  reg2
BP = 949.96, df = 16, p-value < 2.2e-16
```

Figure 5: Breusch-Pagan test on the log-log model.

The test follows a Chi-squared distribution of parameter 16 (17-1=16), whose critical value is 26.296. Since our test is $949.96 > 26.296$ and the p-value is below 5%, I reject the null hypothesis and conclude that there is heteroskedasticty. This means that I cannot rely on the obtained coefficients.

Point b

Is it advisable to use standard errors robust to heteroskedasticity? Report the most appropriate standard error for the coefficient on female.

It is advisable to use robust standard errors for heteroskedasticity because they offer more reliable estimates of the standard errors of the coefficients when homoskedasticity is missing. Therefore, the most appropriate standard error for the coefficient of *female* is the robust one. I calculated it using the *coeftest* command in R.

Here are my results:

```
female       -1.2262e-02  5.3922e-03 -2.2741 0.0229743 *
```

Figure 6: Robust standard errors on the log-log model for female.

The robust standard error for the dummy variable *female* is 0.0053922, whereas in the previous regression it was 0.005426.

Point c

Perform a RESET test to check if the specification is correct. In the test, use up to the third power of the predicted dependent variable.

Since I have found the existence of heteroskedasticity, I would like to know whether it is due to model specification errors or to a real variability of the variance of the errors in a corrected model. To check this aspect I launch a RESET test. The hypotheses are as follows:

$$\begin{cases} H_0 : E[log(y)] = x\beta \\ H_1 : E[log(y)] \neq x\beta \end{cases}$$

To check for this, I used the command *resettest(reg2, power=2:3, type="fitted")* in R:

```
        RESET test

data:  reg2
RESET = 69.548, df1 = 2, df2 = 15227, p-value < 2.2e-16
```

Figure 7: RESET test on the log-log model.

The test follows an F distribution with q, that is equal to 2, and $n - k_{unrestricted}$ $(15246 - 19 = 15227)$ degrees of freedom. The p-value is less than 5%, I reject the null hypothesis and conclude that the specification is wrong, the model hides some important information. The problem should be with the omitted variables or structural breaks.

Point d

Adds to the specification married and if_child. Are the new variables jointly significant?

I created a new regression, called *reg3*, adding two new variables, *married* and *if_child*. The equation then becomes:

$$\log(socladder) = \beta_1 + \beta_2 age + \beta_3 age^2 + \beta_4 female + \beta_5 hi_school + \beta_6 college + \beta_7 empl + \beta_8 self empl + \beta_9 \log(income) + \beta_{10} \log(tot_finwth) + \beta_{11} wave_9 + \beta_{12} wave_10 + \beta_{13} wave_11 + \beta_{14} wave_12 + \beta_{15} wave_13 + + \beta_{16} wave_14 + \beta_{17} wave_15 + \beta_{18} married + \beta_{19} if_child + \epsilon$$

Below are the results in R:

```
Call:
lm(formula = log(socladder) ~ age + I(age^2) + female + hi_school +
    college + empl + selfempl + log(income) + log(tot_finw) +
    wave_9 + wave_10 + wave_11 + wave_12 + wave_13 + wave_14 +
    wave_15 + married + if_child, data = dataset)

Residuals:
     Min       1Q   Median       3Q      Max
-1.97523 -0.12396  0.05531  0.19330  1.11182

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    6.505e-01  1.185e-01   5.489 4.12e-08 ***
age            1.304e-02  3.333e-03   3.914 9.13e-05 ***
I(age^2)      -6.305e-05  2.349e-05  -2.684  0.00727 **
female        -2.027e-03  5.539e-03  -0.366  0.71437
hi_school      6.353e-02  7.571e-03   8.391  < 2e-16 ***
college        1.641e-01  9.074e-03  18.082  < 2e-16 ***
empl           4.541e-02  7.092e-03   6.403 1.57e-10 ***
selfempl       7.486e-02  1.047e-02   7.148 9.18e-13 ***
log(income)    2.856e-02  2.238e-03  12.762  < 2e-16 ***
log(tot_finw)  1.431e-02  6.011e-04  23.812  < 2e-16 ***
wave_9TRUE    -1.009e-02  9.757e-03  -1.034  0.30113
wave_10TRUE   -2.056e-02  9.491e-03  -2.166  0.03035 *
wave_11TRUE   -2.311e-02  9.762e-03  -2.367  0.01793 *
wave_12TRUE   -1.485e-02  9.670e-03  -1.535  0.12476
wave_13TRUE   -2.322e-02  1.059e-02  -2.192  0.02838 *
wave_14TRUE    3.043e-02  1.091e-02   2.788  0.00532 **
wave_15TRUE    3.027e-02  1.178e-02   2.570  0.01017 *
married        5.460e-02  6.072e-03   8.993  < 2e-16 ***
if_child       1.149e-02  1.064e-02   1.080  0.28027
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3243 on 15227 degrees of freedom
Multiple R-squared:  0.1699,    Adjusted R-squared:  0.1689
F-statistic: 173.1 on 18 and 15227 DF,  p-value: < 2.2e-16
```

Figure 8: OLS regression on the log-log model with new variables.

Looking at this table, it can be seen that the explanatory variable *married* is significant. To test the joint significance of the two variables, I launch an F-test with the command *lht*. The following are the hypotheses and results:

$$\begin{cases} H_0 : \beta_{18} = \beta_{19} = 0 \\ H_1 : \beta_{18} \neq \beta_{19} \neq 0 \end{cases}$$

```
> lht(reg3, c("married=0", "if_child=0" ), test = "Chisq")
Linear hypothesis test

Hypothesis:
married = 0
if_child = 0

Model 1: restricted model
Model 2: log(socladder) ~ age + I(age^2) + female + hi_school + college +
    empl + selfempl + log(income) + log(tot_finw) + wave_9 +
    wave_10 + wave_11 + wave_12 + wave_13 + wave_14 + wave_15 +
    married + if_child

  Res.Df    RSS Df Sum of Sq  Chisq Pr(>Chisq)
1  15229 1610.2
2  15227 1601.0  2    9.2859 88.321  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 9: F-test on the new variables.

The test follows a Chi-squared distribution of parameter 2, whose critical value is 5.991. Since $88.321 > 5.991$ and the p-value is under 5%, I reject the null hypothesis and conclude that the two variables are jointly significant.

Regression Model Analysis: Choosing Between PE, RE, and FE

Point a

Run a test to check if we should prefer PE or RE.

As requested I focus on the baseline regression taking as dependent variable the log transformation of *socladder*.
To check which is more preferable between the pooled effect and the random effect, I should first estimate both PE (*reg4*) and RE (*reg5*), using the *plm* command in R. Then, to check which model I should prefer, I applied the Breusch-Pagan test, using the command *plmtest(reg4, type=c("bp"))*.
The set of hypothesis is as follows:

$$\begin{cases} H_0 : \sigma_\alpha^2 = 0 \\ H_1 : \sigma_\alpha^2 \neq 0 \end{cases}$$

The result of the test is:

```
        Lagrange Multiplier Test - (Breusch-Pagan)

data:  log(socladder) ~ age + I(age^2) + female + hi_school + college +  ...
chisq = 612.45, df = 1, p-value < 2.2e-16
alternative hypothesis: significant effects
```

Figure 10: Breusch-Pagan test: PE vs RE.

The test follows a Chi-squared distribution of parameter 1, whose critical value is 3.841. Since $612.45 > 3.841$ and the p-value is under the 5%, I reject the null hypothesis and conclude that RE is preferred to PE.

Point b

Run a test to check if we should prefer PE or FE.

Since I have already computed the PE (*reg4*), in this section I just computed FE (*reg6*) by using the command *plm* and specifying in the section "*model = within*". To check which model I should prefer I applied a Chow test. The set of hypothesis is as follows:

$$\begin{cases} H_0 : \alpha_i = 0 \\ H_1 : \alpha_i \neq 0 \end{cases}$$

In R I used the command *pFtest(reg6, reg4)*. Here below my results:

```
        F test for individual effects

data:  log(socladder) ~ age + I(age^2) + female + hi_school + college +  ...
F = 2.3903, df1 = 11284, df2 = 3945, p-value < 2.2e-16
alternative hypothesis: significant effects
```

Figure 11: F-test: PE vs FE.

The test follows a Chi-squared distribution of parameter *N-1*. The p-value is under the 5%, thus I reject the null hypothesis and conclude that FE is preferred to PE.

Point c

Run a test to check if we should prefer RE or FE.

To test which model is the best between RE and FE, I should perform a Hausman test, whose set of assumptions is defined as follows:

$$\begin{cases} H_0 : E[\alpha_i|x] = 0 \\ H_1 : E[\alpha_i|x] \neq 0 \end{cases}$$

The results are the following:

```
        Hausman Test

data:  log(socladder) ~ age + I(age^2) + female + hi_school + college +  ...
chisq = 140.16, df = 13, p-value < 2.2e-16
alternative hypothesis: one model is inconsistent
```

Figure 12: Hausman test: RE vs FE.

The test follows a Chi-squared distribution of parameter *k*. Since The p-value is under the 5%, I reject the null hypothesis and conclude that FE is preferred to RE.

Point d

Choose the best model and report the effect of increasing financial wealth by 10%. Test if the effect is significantly different from zero.

Since it appeared in the previous point that the fixed effects estimator is preferred over the others, I chose it as the best model. I run the FE model represented by *reg6* and this is the result for the financial wealth variable:

```
log(tot_finw)  3.2752e-03  1.2145e-03  2.6967 0.0070331 **
```

Figure 13: FE on the variable financial wealth

An increase of 10% of *financial wealth*, creates an increase of 0,032752% on the dependent variable *self-assessed social status*. To test its significance, I check the t-value, which is greater than 1.96 ($2.6967 > 1.96$) and thus I reject the null hypothesis and conclude that the coefficient is significant.

Analysis of Social Status Determinants: LPM and Probit Regression

Point a

Report the probability of assessing high social status for a female aged 55, employee with high school education, earning 30 thousand USD, with wealth 80 thousand USD and observed in wave 12.

I created the dummy variable, that is equal to 1 when *socladder* is above 5 and it is equal to 0 in the other cases. I performed the baseline regression, using as dependent variable *socladder_dummy*. I called this new regression *reg7*. As it is a LPM regression, there is heteroskedasticity, thus I attached to the coefficients estimates the robust standard errors. Here are my results:

```
> coeftest(reg7, vcov = vcovHC(reg7, type ="HC1"))

t test of coefficients:

                Estimate  Std. Error t value  Pr(>|t|)
(Intercept)   -3.6706e-01  1.5588e-01 -2.3547 0.0185486 *
age            1.1110e-02  4.2892e-03  2.5903 0.0095991 **
I(age^2)      -5.4644e-05  3.0058e-05 -1.8179 0.0690933 .
female        -1.5486e-02  6.9066e-03 -2.2422 0.0249624 *
hi_school      7.5686e-02  1.1095e-02  6.8216 9.345e-12 ***
college        1.9201e-01  1.2099e-02 15.8704 < 2.2e-16 ***
empl           3.0774e-02  9.3755e-03  3.2824 0.0010316 **
selfempl       7.1288e-02  1.2008e-02  5.9366 2.973e-09 ***
log(income)    3.5508e-02  3.4633e-03 10.2527 < 2.2e-16 ***
log(tot_finw)  1.4958e-02  8.1939e-04 18.2551 < 2.2e-16 ***
wave_9TRUE    -3.3607e-02  1.2404e-02 -2.7093 0.0067492 **
wave_10TRUE   -3.2752e-02  1.2083e-02 -2.7106 0.0067241 **
wave_11TRUE   -4.1219e-02  1.2516e-02 -3.2933 0.0009924 ***
wave_12TRUE   -2.5480e-02  1.2274e-02 -2.0760 0.0379143 *
wave_13TRUE   -3.8081e-02  1.3573e-02 -2.8056 0.0050283 **
wave_14TRUE    9.7316e-03  1.3387e-02  0.7269 0.4672699
wave_15TRUE    2.7795e-02  1.4309e-02  1.9424 0.0521012 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 14: LPM with robust standard errors

The probability is given by the linear combination of the coeffieicent estimates, each multiplied by the value of the explanatory variable requested. This is the result:

```
> fit1 = coef(reg7)["(Intercept)"] +coef(reg7)["female"]*1 +coef(reg7)["age"]*55 +
coef(reg7)["hi_school"]*1 +coef(reg7)["empl"]*1 +coef(reg7)["log(income)"]*log(300
00) +coef(reg7)["log(tot_finw)"]*log(80000) +coef(reg7)["wave_12TRUE"]*1
> fit1 = pnorm(fit1, mean=0, sd=1)
> fit1
(Intercept)
  0.8007814
```

Figure 15: Probability of assessing high social status.

An individual with these characteristics has a 80.07% probability of assessing high social status.

Point b

Test if the effect of empl and selfempl are equal.

To test whether the effects of *empl* and *selfempl* are equal, I run an F-test, using the command *lht*. The following are the hypotheses and results:

$$\begin{cases} H_0 : \beta_7 = \beta_8 \\ H_1 : \beta_7 \neq \beta_8 \end{cases}$$

```
> lht(reg7, c("empl=selfempl"), test="Chisq")
Linear hypothesis test

Hypothesis:
empl - selfempl = 0

Model 1: restricted model
Model 2: socladder_dummy ~ age + I(age^2) + female + hi_school + college +
    empl + selfempl + log(income) + log(tot_finw) + wave_9 +
    wave_10 + wave_11 + wave_12 + wave_13 + wave_14 + wave_15

  Res.Df    RSS Df Sum of Sq  Chisq Pr(>Chisq)
1  15230 2648.2
2  15229 2646.8  1    1.4682 8.4475   0.003655 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 16: F-test between empl and selfempl.

The test follows a Chi-squared distribution of parameter 1, whose critical value is 3.841. Since $8.4475 > 3.841$ and the p-value is lower than 5%, I reject the null hypothesis and conclude that the effect of *empl* and *selfempl* is not equal.

Point c

Repeat the model with a probit regression. Test if the effect of empl and selfempl are equal.

To run the PROBIT regression in R, I applied the command *glm*. Below is the result:

```
Call:
glm(formula = socladder_dummy ~ age + I(age^2) + female + hi_school +
    college + empl + selfempl + log(income) + log(tot_finw) +
    wave_9 + wave_10 + wave_11 + wave_12 + wave_13 + wave_14 +
    wave_15, family = binomial(link = "probit"), data = dataset)

Coefficients:
               Estimate Std. Error z value Pr(>|z|)
(Intercept)   -2.6304665  0.5126986  -5.131 2.89e-07 ***
age            0.0324276  0.0144205   2.249  0.02453 *
I(age^2)      -0.0001558  0.0001018  -1.530  0.12599
female        -0.0646881  0.0237312  -2.726  0.00641 **
hi_school      0.1917382  0.0305101   6.284 3.29e-10 ***
college        0.6962888  0.0399398  17.433  < 2e-16 ***
empl           0.0850637  0.0307264   2.768  0.00563 **
selfempl       0.2780247  0.0501152   5.548 2.89e-08 ***
log(income)    0.1095233  0.0097495  11.234  < 2e-16 ***
log(tot_finw)  0.0478774  0.0025088  19.084  < 2e-16 ***
wave_9TRUE    -0.1177497  0.0426639  -2.760  0.00578 **
wave_10TRUE   -0.1122489  0.0415444  -2.702  0.00689 **
wave_11TRUE   -0.1367743  0.0423810  -3.227  0.00125 **
wave_12TRUE   -0.0866688  0.0423155  -2.048  0.04054 *
wave_13TRUE   -0.1232110  0.0462802  -2.662  0.00776 **
wave_14TRUE    0.0404263  0.0489970   0.825  0.40933
wave_15TRUE    0.1123329  0.0537309   2.091  0.03656 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 17772  on 15245  degrees of freedom
Residual deviance: 15861  on 15229  degrees of freedom
AIC: 15895

Number of Fisher Scoring iterations: 5
```

Figure 17: PROBIT regression.

To check whether the effect of *empl* and *selfempl* is equal, I repeat the same command as in step b, using the new PROBIT regression. I run an F-test, using the command *lht*. The set of hypotheses is as follows:

$$\begin{cases} H_0 : \beta_7 = \beta_8 \\ H_1 : \beta_7 \neq \beta_8 \end{cases}$$

Here below my result:

```
> lht(reg8, c("empl=selfempl"), test="Chisq")
Linear hypothesis test

Hypothesis:
empl - selfempl = 0

Model 1: restricted model
Model 2: socladder_dummy ~ age + I(age^2) + female + hi_school + college +
    empl + selfempl + log(income) + log(tot_finw) + wave_9 +
    wave_10 + wave_11 + wave_12 + wave_13 + wave_14 + wave_15

  Res.Df Df  Chisq Pr(>Chisq)
1  15230
2  15229  1 13.986  0.0001842 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 18: F-test between empl and selfempl under PROBIT regression.

The test follows a Chi-squared distribution of parameter 1, whose critical value is 3.841. Since $13.986 > 3.841$ and the p-value is lower than 5%, I reject the null hypothesis and conclude that the effect of *empl* and *selfempl* is different.

Point d

Report the percentage of observations on which the probit model can correctly predict the dependent variable.

To know the percentage of observations over which the PROBIT model can correctly predict the dependent variable, I calculated the Count-R^2 statistic, using the command *CountR2* in R. Here is my result below:

```
> CountR2 =function(m) mean(m$y==round(m$fitted.values))
> CountR2(reg8)
[1] 0.7377673
```

Figure 19: Count-R^2.

The Count-R^2 statistic provides indications of the goodness of fit. In this case the fit is good because the predictions are correct in 73.77% of the observations.